David Fernandez

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Summary

I am a Ph.D. candidate in Computer Science at Clemson University, bridging explainable AI, agentic systems, and security for safety-critical applications. My work on LLMs and VLMs has appeared at venues including WACV, DSN, IEEE IV, CVPR Workshops, and USENIX VehicleSec (six first-authored papers), advancing component-level explainability, zero-shot reasoning, and adversarial scenario analysis. Through Clemson's VIPR-GS program, I develop LLM reasoning frameworks and VLM evaluation systems for the U.S. Army's Next Generation Combat Vehicle (NGCV) program. At BMW Group, I design agentic AI systems, including autonomous prompt optimization pipelines and defenses against multi-turn adversarial attacks.

Education

2024 - present	Ph.D. in Computer Science at **Clemson University**	(GPA: 4.0/4.0)
2022 - 2024	M.S. in Data Science and Informatics at **Clemson University**	(GPA: 4.0/4.0)
2010 - 2018	Dual B.S. in Computer Engineering & Business Administration at **ITAM**	(Mexico City)

Work Experience

Virtual Prototyping of Ground Systems (VIPR-GS) for U.S. Army's NGCV program, Clemson University 2025 - Present

- Developed perception model architectures using semantic embeddings to bridge visual scene understanding with regulatory compliance, enabling component-level diagnostics in autonomous systems
- Built novel multimodal fusion architectures combining camera and sensor data, achieving quantifiable improvements in perception accuracy for safety-critical scenarios
- Integrated VLMs with high-fidelity simulations to evaluate multimodal decision-making
- Created Meta-VLM (WACV 2027), a framework for evaluating metadata in VLM driving decisions, identifying a minimal metadata subset that retains 92.9% of full-metadata performance with 43% fewer tokens and 9% lower inference latency
- Integrated VLMs with semantic embeddings for zero-shot reasoning, achieving a 7% improvement over deep learning baselines

AI Researcher at BMW Group 2024 - Present

- Co-authored SACAIR 2025 publication on edge AI deployment, implementing small language model framework enabling distributed inference on constrained devices
- Designed multilingual semantic document framework using transformer embeddings to automate compliance verification, detecting structural and semantic changes across regulatory documents
- Conducted systematic security analysis of production LLM systems through 10,000+ multi-turn adversarial interactions, discovering critical prompt injection vulnerabilities
- Architected novel context-aware defense framework addressing fundamental limitations in existing industry solutions, researching new security paradigm for enterprise conversational AI systems

PhD Research Assistant at Clemson University 2024 - Present

- Published six first-authored papers advancing explainable AI through collaborative research with Army VIPR and BMW, bridging academic work with real-world deployment in safety-critical AI systems
- Teaching Assistant for "Security in Emerging Computing and Networking Systems," delivering lectures, mentoring students on semester-long research projects in adversarial ML and security topics

Lead Engineer & Technical Architect at MUSEOLAB 2016 - 2021

– Designed and deployed 5 production ML pipelines with ensemble methods and clustering algorithms, establishing foundation in model development, optimization, and real-world deployment
– Led full-stack development of ML-integrated systems, including model training, optimization, and production deployment workflows

Awards & Honors

2026 **SAE International Award for Outstanding Student Paper honoring Phil Myers**
For "Avoiding the Crash: A Vision-Language Model Evaluation of Critical Traffic Scenarios"

Publications

Fernandez, **David**, Pedram MohajerAnsari, Amir Salarpour, and Mert D. Pesé (2027). "Meta-VLM: Evaluating Metadata for Vision-Language Driving Decisions in Safety-Critical Scenarios". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. To appear.

Fernandez, **David**, Pedram MohajerAnsari, Cigdem Kokenoz, et al. (2026a). "From MIRAGE to CLEAR: Component-Level Explainable Anomaly Reasoning for Autonomous Vehicle Perception Systems". In: *International Conference on Dependable Systems and Networks (DSN)*.

Fernandez, **David**, Pedram MohajerAnsari, Amir Salarpour, and Mert D. Pesé (2026b). "Understanding Adversarial Transferability in Vision-Language Models for Autonomous Driving: A Cross-Architecture Analysis". In: *SAE Technical Paper 2026-01-0170*. DOI: 10.4271/2026-01-0170.

Fernandez, **David**, Pedram MohajerAnsari, Amir Salarpour, et al. (2026c). "Comparative Analysis of Patch Attack on VLM-Based Autonomous Driving Architectures". In: *2026 IEEE Intelligent Vehicles Symposium (IV)*.

MohajerAnsari, Pedram, Amir Salarpour, **David Fernandez**, and Mert D. Pesé (2026). "Budget-Aware Adaptive Adversarial Patches for Black-Box Object Detection". In: *2026 IEEE International Conference on Image Processing (ICIP)*.

Salarpour, Amir, Pedram MohajerAnsari, **David Fernandez**, et al. (2026). "Trustworthy AI for CPS Security". In: *AI for CPS Security and Resilience*. To appear. Springer.

Duffy, Edward B., **David Fernandez**, Alta de Waal, and Mert D. Pesé (2025). "Small Language Models on the Edge for Real-World Agentic Systems in Industry". In: *Southern African Conference for Artificial Intelligence Research (SACAIR)*.

Fernandez, **David**, Pedram MohajerAnsari, Cigdem Kokenoz, et al. (2025a). "WIP: From Detection to Explanation: Using LLMs for Adversarial Scenario Analysis in Vehicles". In: *3rd USENIX Symposium on Vehicle Security and Privacy (VehicleSec '25)*.

Fernandez, **David**, Pedram MohajerAnsari, Amir Salarpour, and Mert D. Pesé (2025b). "Avoiding the Crash: A Vision-Language Model Evaluation of Critical Traffic Scenarios". In: *SAE Technical Paper 2025-01-8213*. DOI: 10.4271/2025-01-8213.

MohajerAnsari, Pedram, Amir Salarpour, **David Fernandez**, et al. (2025). "Attention-Aware Temporal Adversarial Shadows on Traffic Sign Sequences". In: *IEEE/CVF CVPR Workshops*, pp. 3600–3608.

Salarpour, Amir, Pedram MohajerAnsari, **David Fernandez**, and Mert D. Pesé (2025). "SASA: Sequence-Aware Shadow Attacks via Attention Alignment for Traffic Sign Recognition". In: *The 6th Workshop of Adversarial Machine Learning on Computer Vision: Safety of Vision-Language Agents*.

Skills

Programming Languages	Python, Java, Swift, JavaScript, Shell Scripting, HTML
Libraries / Architectures	PyTorch, CUDA, scikit-learn, Transformers, Vision Transformers, CNNs
Computer Vision & ML	Object Detection, Segmentation, Multimodal Fusion, Model Optimization
LLMs & Agents	Agentic Systems, Prompt Optimization, Adversarial Defense, Edge AI